Department of Electrical and Computer Engineering
Utah State University
ECE 5480/6480: VLSI Testing and Verification
Instructor: Dr. Roy
Hardware Security Project-Research Based

Assigned Date: Check Schedule
Due Date: Check Schedule

Introduction

This project is more open ended and can be chosen as an alternate option to the fault detection/diagnosis project. The goal of this project is to understand some concepts about Hardware Trojan and then detect such Trojans in a few circuits from the Trust-HUB benchmark suite.

The goal of this lab is to generate an economical test set for a given circuit, and then implement a fault diagnosis procedure on a given netlist to identify faults. Producing an economical set of test-vectors will reduce the lifecycle cost. An economical set will eliminate less functional test vectors, while simultaneously increasing fault/defect coverage. This allows for improved time to revenue for a given circuit. The most important utility of the test set is reduction in order to reduce the per chip testing time for automatic test equipment. In logic diagnosis, fault models (e.g. stuckat fault, bridge fault, open fault, and etc.) are used to model failure behavior caused by a physical defect. Using a logic level fault model can simplify simulating the fault effects caused by a real defect. When a chip fails at testing, fault diagnosis can determine the most likely faulty locations and fault types. Fault diagnosis focuses on defects that are present on the pins of a library cell or the interconnecting wires between library cells.

PROCEDURE:

1. Generate an Economical Test Set

Develop an economical test set which has sufficient test and fault coverage and has somewhat minimum size for a given circuit using the TetraMAX tool. Each person will be provided their own circuits. All these circuits are developed to have similar test generation complexity. You can generate test vectors using ATPG tool, or generate random test vectors. Feel free to leverage various techniques such as combination, compaction, replacement, etc. to create an economical test set. Note that the required fault coverage for this test set is at least 90%.

2. Perform Fault Diagnosis

Three netlists which are realizations of the circuit for which you have generated test vectors are provided. Each of these netlists are Hypothetical Fabricated Chips (HFC). Test each HFC with your economical test set to identify whether the HFC is faulty or faultfree. One or more of the given HFCs are faulty. After determining which HFC(s) are faulty, choose one (if you find more than one faulty HFC) and perform fault diagnosis to identify the actual fault in that HFC. Because the fault, which is in the faulty HFC, may be equivalent to other faults and collapsed by them, your test set might not be able to excite the fault. In this case, you need to find the desired test vector for exciting the fault and add it to your test set.

Diagnosis hints:

1. Labs + TetraMAX user guide.
2. Building your own tool/program to diagnose the faulty circuits.
3. After finding possible faults in the faulty HFC, you can inject each of them in the good circuit and compare the results. Based on the comparison, you will be able to identify the actual fault.

It should be said that these technique are just suggestions. You are allowed to figure out and implement other useful techniques as well.

DELIVERABLES:

You must submit the following items:

1) Report:

I. **Result**: Please include all the reports and results that you obtain while completing each part (such as total number of test vectors in your test set, test report and fault coverage, total number of detected faults and undetected/undetectable faults, all simulation results for the given circuit and HFCs). Also, you should include results that were obtained by performing fault diagnosis on a chosen faulty HFC.

II. **Discussion**: This section needs to contain the initial strategy and techniques that you planned to do each part, as well as the final strategy and techniques that you used. Provide evidence showing the effectiveness of your strategies and techniques to us. Also, provide details of the process that you used for each part (e.g. for generating economical test set, What steps were used, and how did these steps reduce the number of test vectors or increase fault coverage?) Create a flowchart explaining your final strategy for generating an economical test set. Also create a flowchart explaining your final strategy for performing a fault diagnosis procedure. Finally, please explain the problem(s) that you faced in each part and your solution(s) for them as well as suggest and discuss other strategies that can be used for each part.

2) Files: Include all programs and scripts that you created, as well as any of the netlists that you modified. Provide an explanation for each of them.

Good Luck